

Операционные системы

Командная оболочка Midnight Commander

Философова Милана Робертовна

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

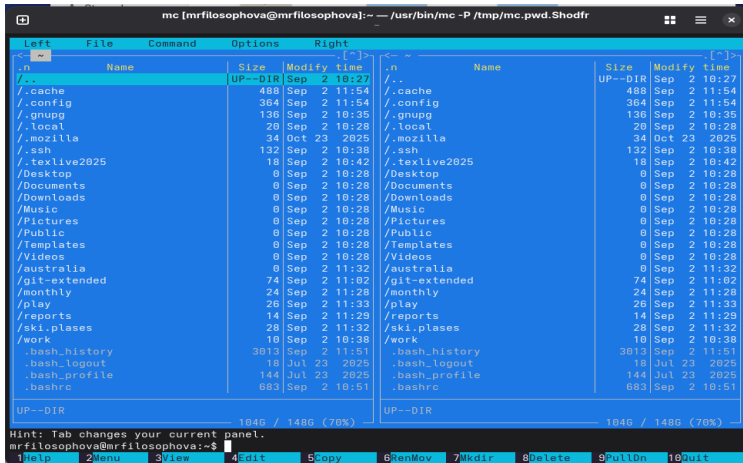


Рисунок 1: Запуск ms

Работа с Midnight Commander

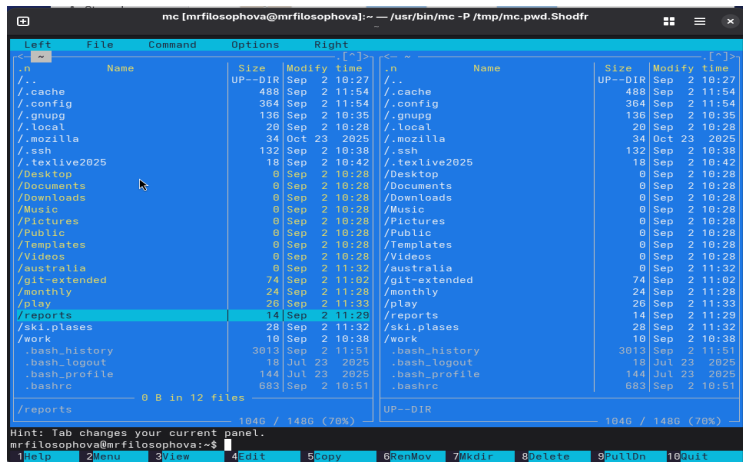


Рисунок 2: Выделение

Работа с Midnight Commander

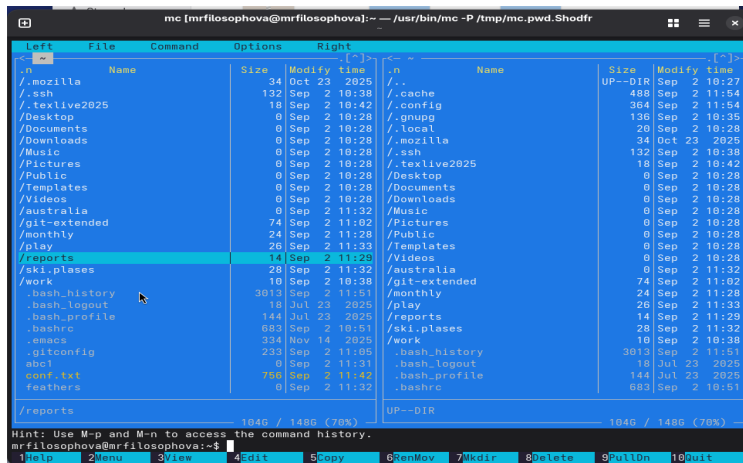


Рисунок 3: Отмена

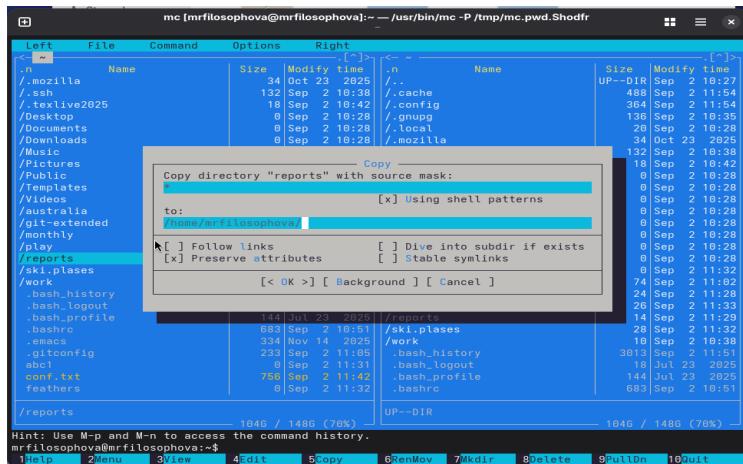


Рисунок 4: Копирование

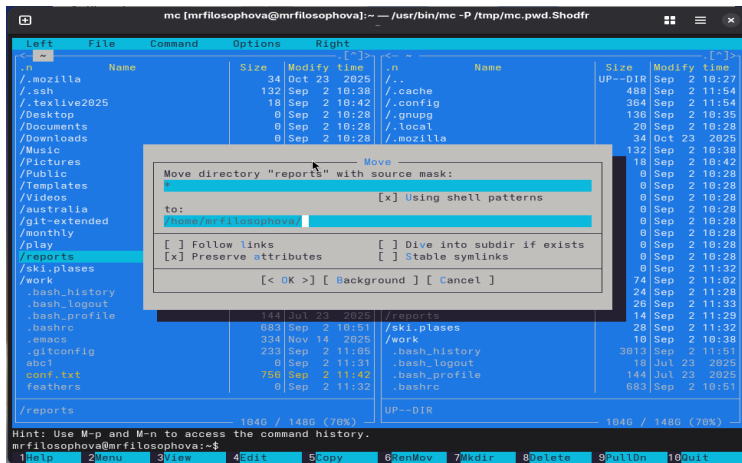


Рисунок 5: Перемещение

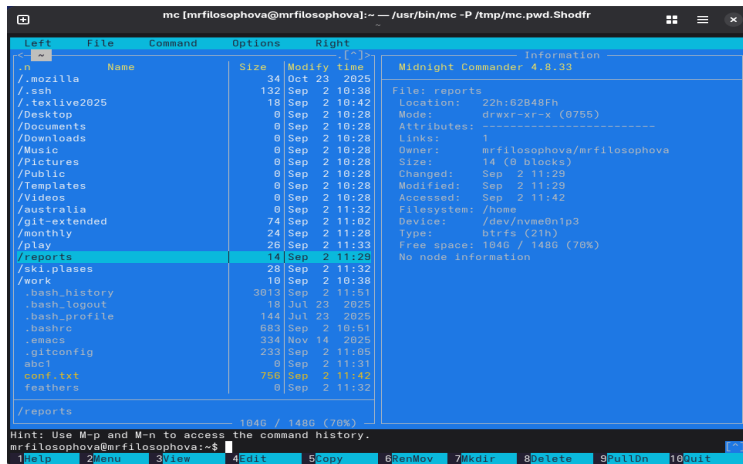


Рисунок 6: Информация

Работа с Midnight Commander

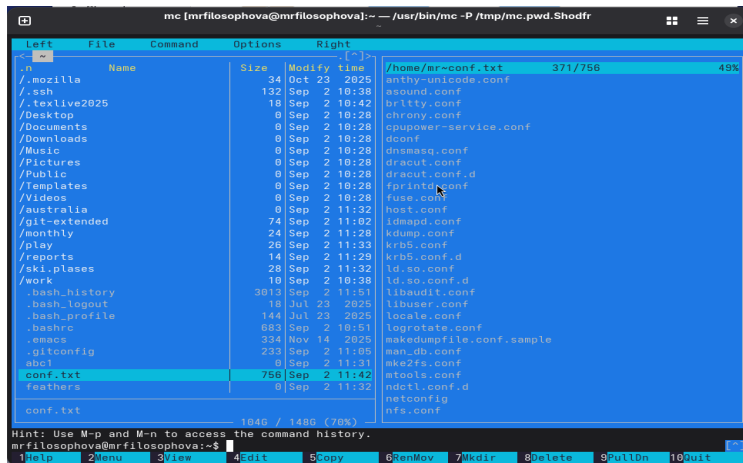


Рисунок 7: Быстрый просмотр

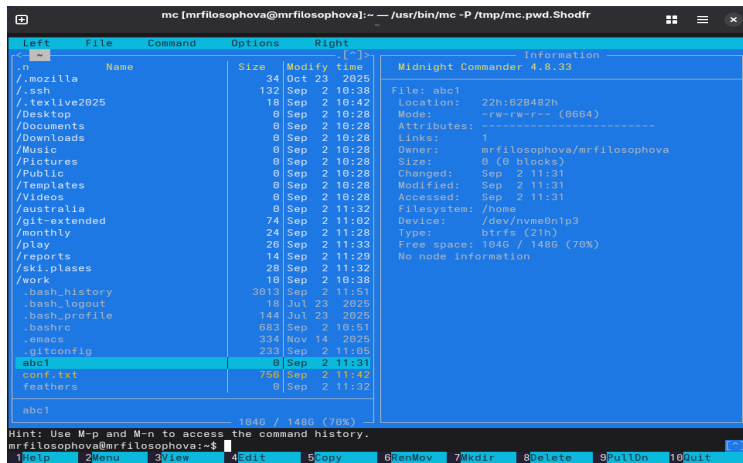


Рисунок 8: Информация

Работа с Midnight Commander

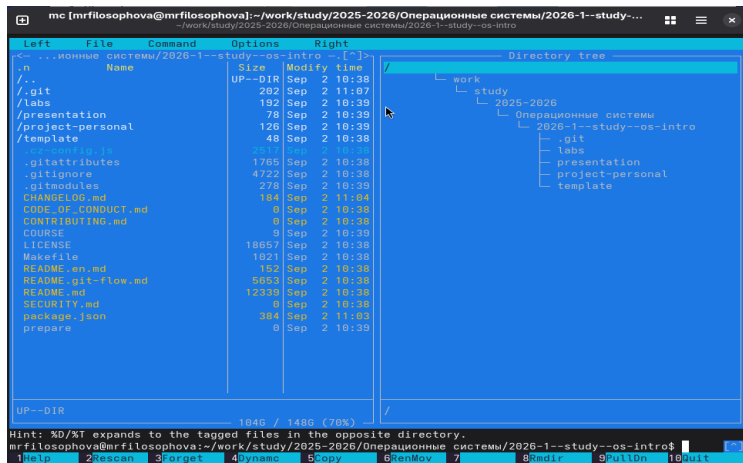
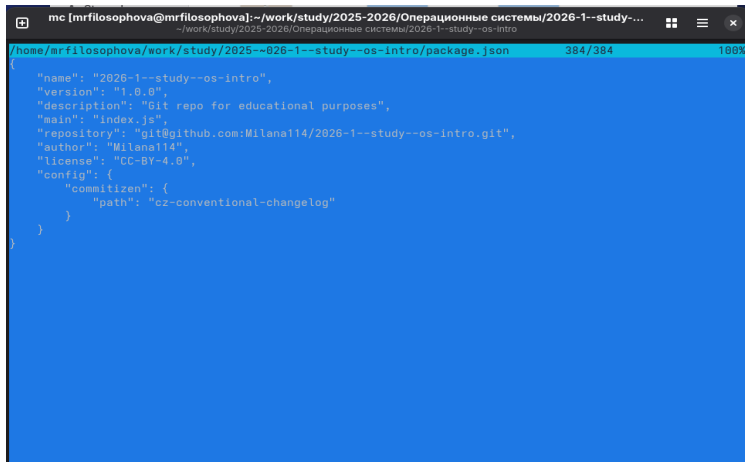


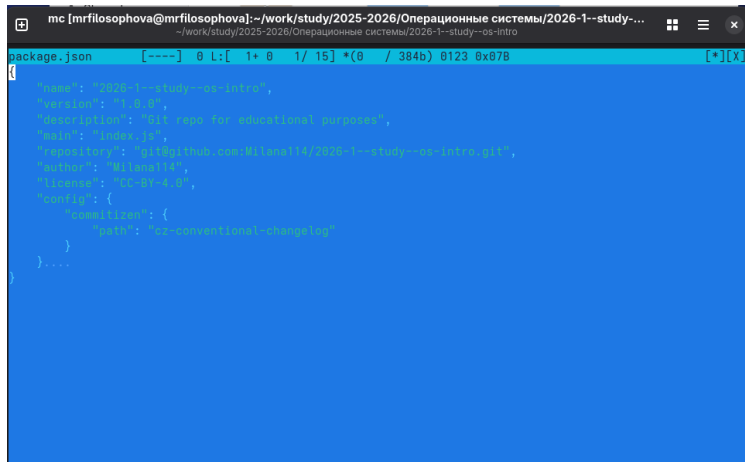
Рисунок 9: Дерево каталогов



The screenshot shows a terminal window of Midnight Commander. The title bar indicates the user is 'mrfilosophova' and the current directory is '~/work/study/2025-2026/Операционные системы/2026-1--study-...'. The main pane displays the contents of the file 'package.json' located at '/home/mrfilosophova/work/study/2025--826-1--study--os-intro/package.json'. The file size is 384/384 bytes and it is 100% visible. The content of the file is a JSON object with the following fields: 'name', 'version', 'description', 'main', 'repository', 'author', 'license', and 'config'. The 'config' field contains a 'commitizen' object with a 'path' field.

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:Milana114/2026-1--study--os-intro.git",
  "author": "Milana114",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 10: Просмотр содержимого текстового файла



The screenshot shows the Midnight Commander (mc) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The file `package.json` is open in the editor. The file content is a JSON object representing a project configuration. The editor has a blue background and a light blue header bar. The file status bar at the top shows `package.json` with various attributes: `[----] 0 L:[ 1+ 0 1/ 15] *(0 / 384b) 0123 0x07B`. The JSON content is as follows:

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:Milana114/2026-1--study--os-intro.git",
  "author": "Milana114",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

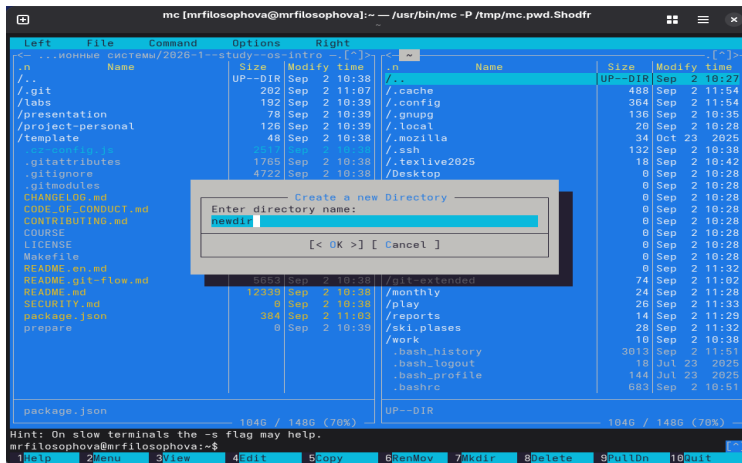


Рисунок 12: Создание каталога

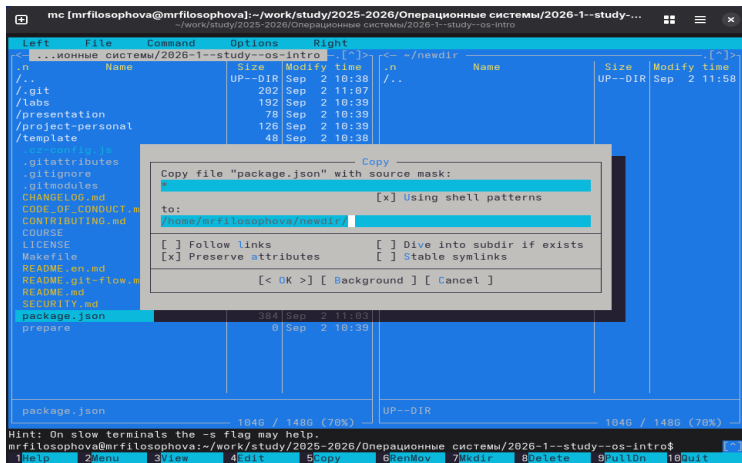


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

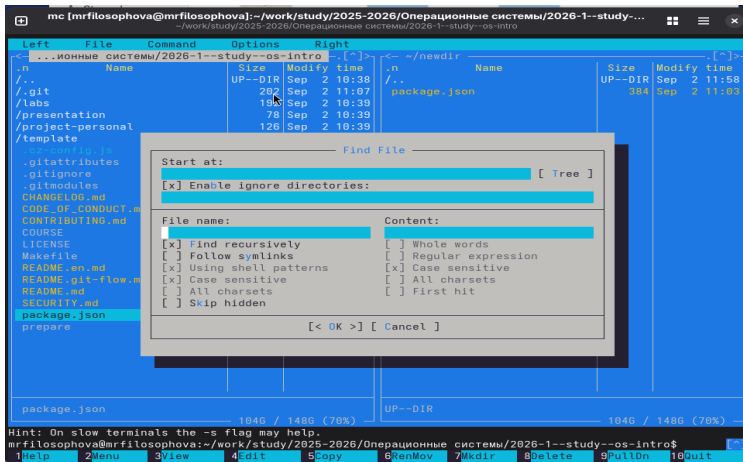


Рисунок 14: Поиск файлов

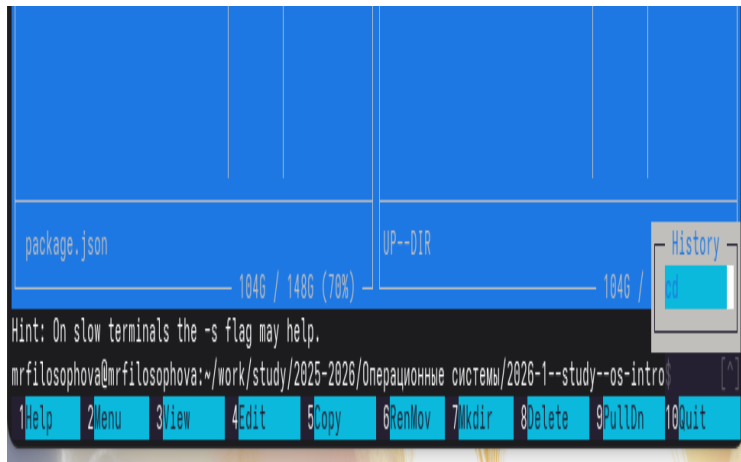


Рисунок 15: История команд

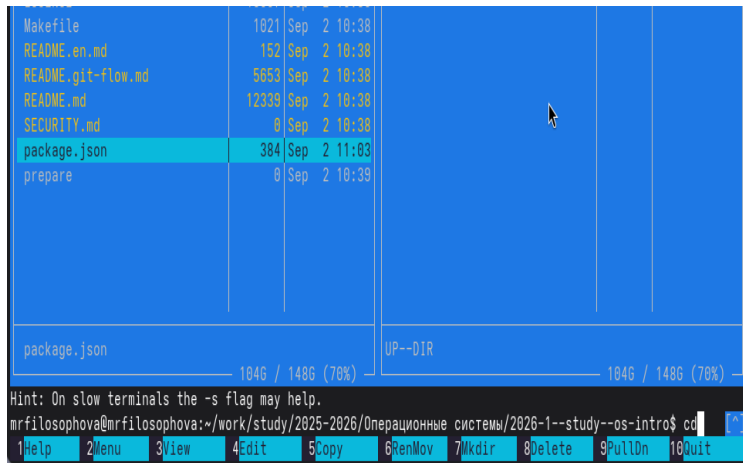
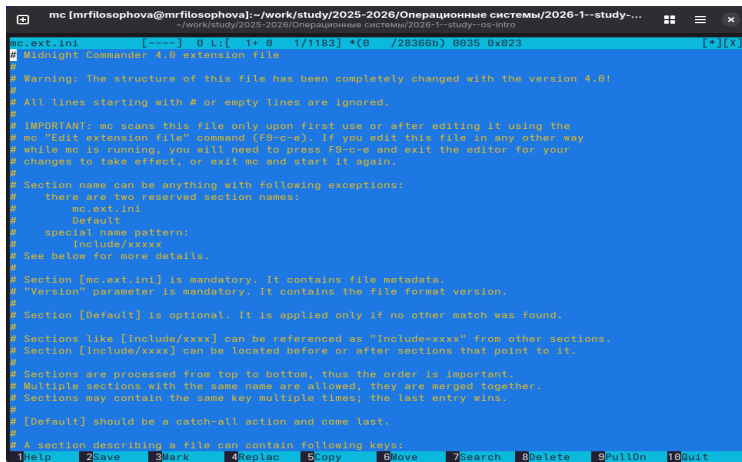


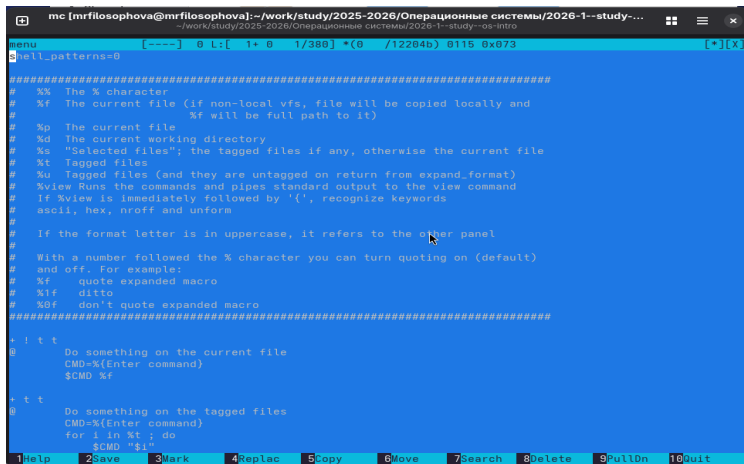
Рисунок 16: Переход в домашний каталог



The screenshot shows a terminal window of the Midnight Commander file manager. The title bar indicates the user is 'mrfilosophova' and the current directory is '~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro'. The main window displays the contents of the 'mc.ext.ini' file, which is a configuration file for Midnight Commander. The file content includes a warning about version 4.0 changes, instructions on how to edit the file, and a list of section names and their purposes. At the bottom of the terminal, there is a status bar with numbered shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, and 10Quit.

```
mc [mrfilosophova@mrfilosophova]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
mc.ext.ini [----] 0 L: [ 1+ 0 1/1183] *(0 /28366b) 8035 8x023 [*)(X]
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT! mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [include/xxxx] can be referenced as "include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
#
# A section describing a file can contain following keys:
```

Рисунок 17: Просмотр файла расширений

A screenshot of the Midnight Commander (MC) application window. The window title is "mc [mrilosophova@mrilosophova]:~/work/study/2025-2026/Операционные системы/2026-1--study-...". The main content area has a blue background and displays the "File" menu. The menu options are: "menu", "shell_patterns=0", a separator line, a list of format codes (%%, %f, %p, %d, %s, %t, %u, %view) with their descriptions, another separator line, and two macros: "+ ! t t" (Do something on the current file) and "+ t t" (Do something on the tagged files). At the bottom, there is a status bar with numbered shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

```
mc [mrilosophova@mrilosophova]:~/work/study/2025-2026/Операционные системы/2026-1--study-...
~/.work/study/2025-2026/Операционные системы/2026-1--study--os-intro

menu [----] 0 L:[ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [+][X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
# %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
# ascii, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %if ditto
# %0f don't quote expanded macro
#####

+ ! t t
@ Do something on the current file
  CMD=(Enter command)
  $CMD %f

+ t t
@ Do something on the tagged files
  CMD=(Enter command)
  for i in %t ; do
    $CMD "$i"

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 18: Просмотр файла меню

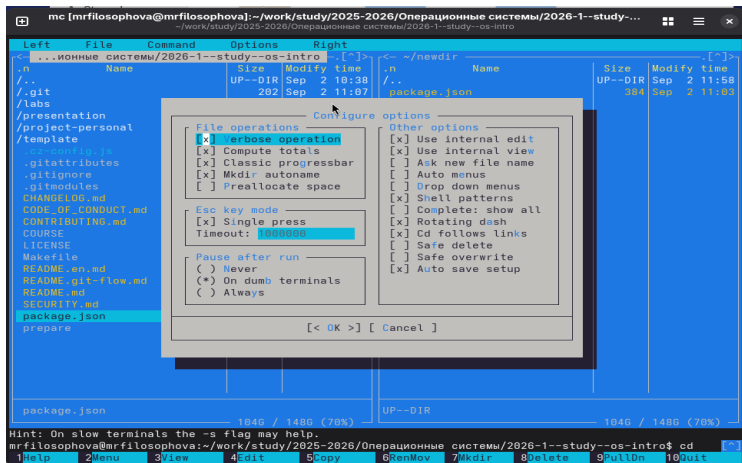


Рисунок 19: Конфигурация

Работа с Midnight Commander

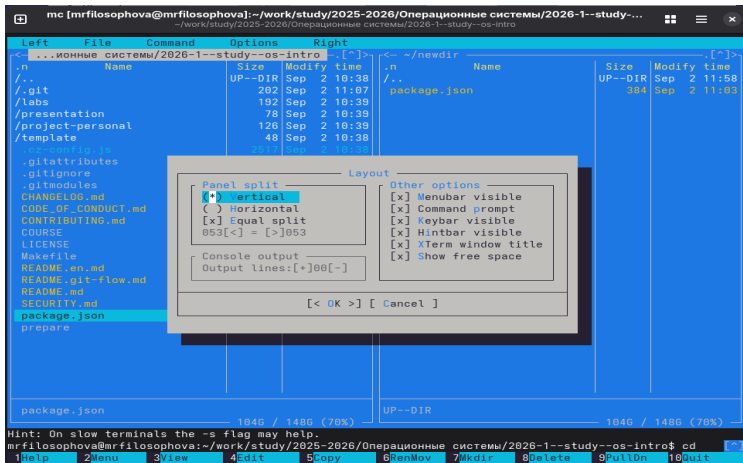


Рисунок 20: Внешний вид

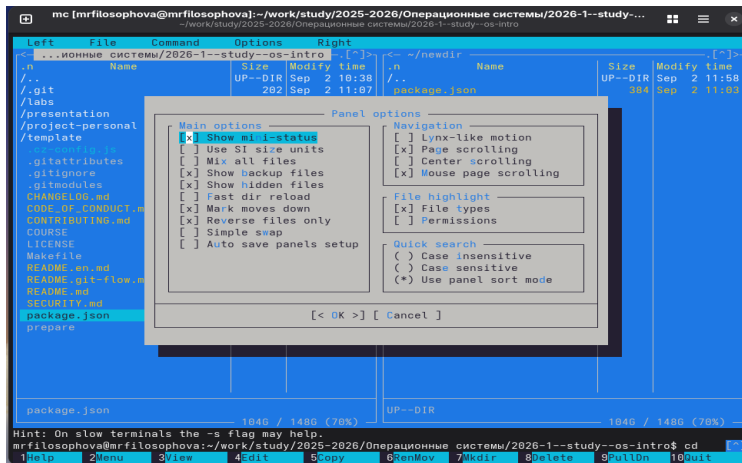


Рисунок 21: Настройки панелей

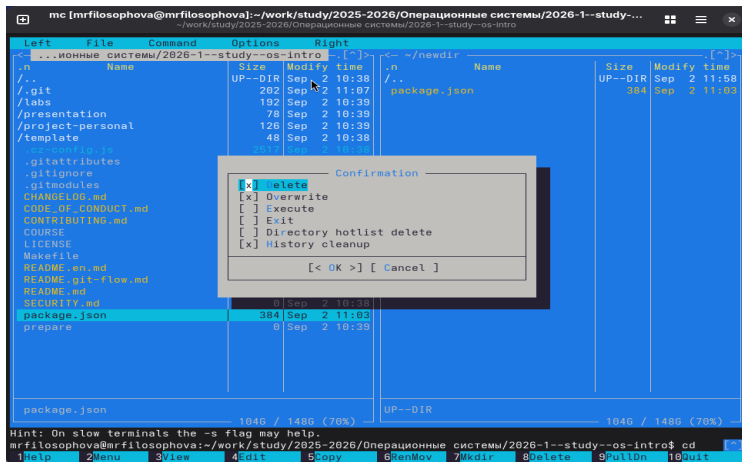


Рисунок 22: Подтверждение

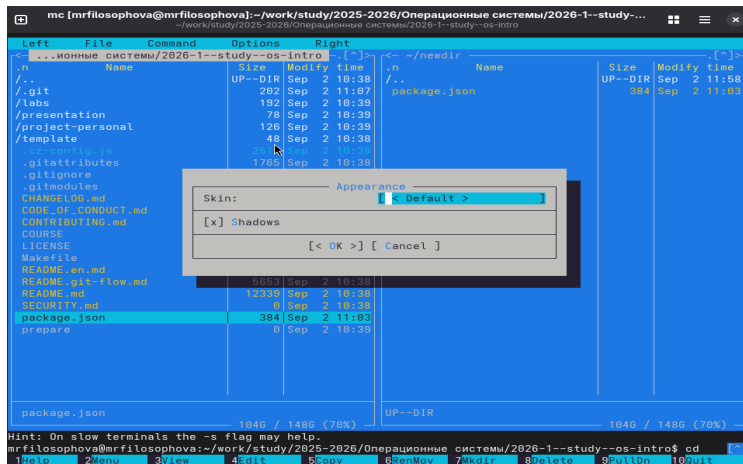


Рисунок 23: Оформление

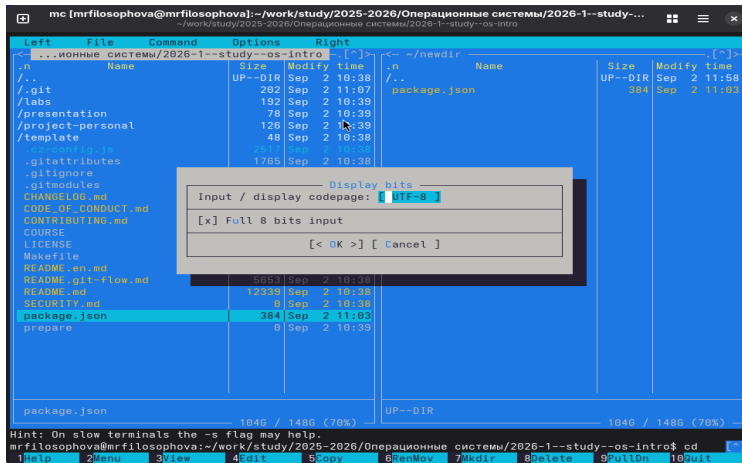


Рисунок 24: Кодировка символов

Работа с Midnight Commander

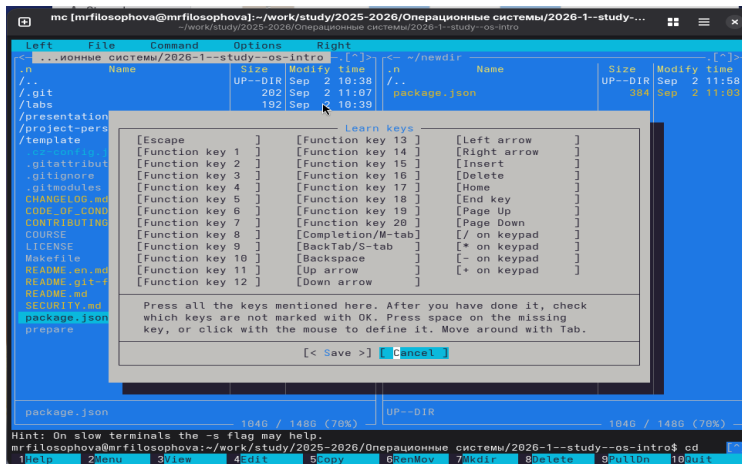
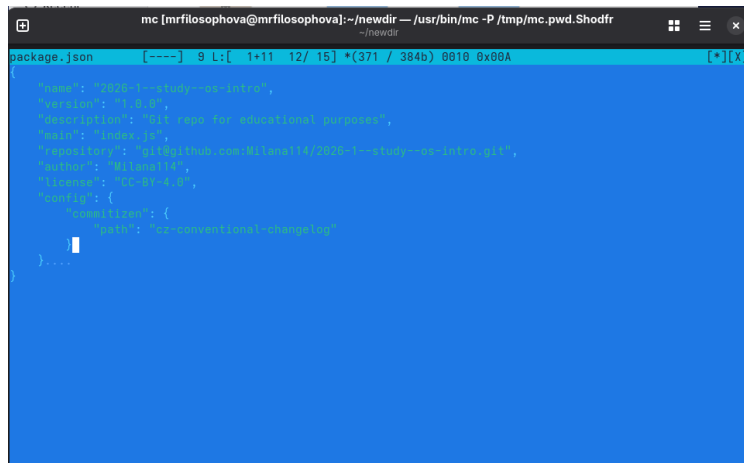


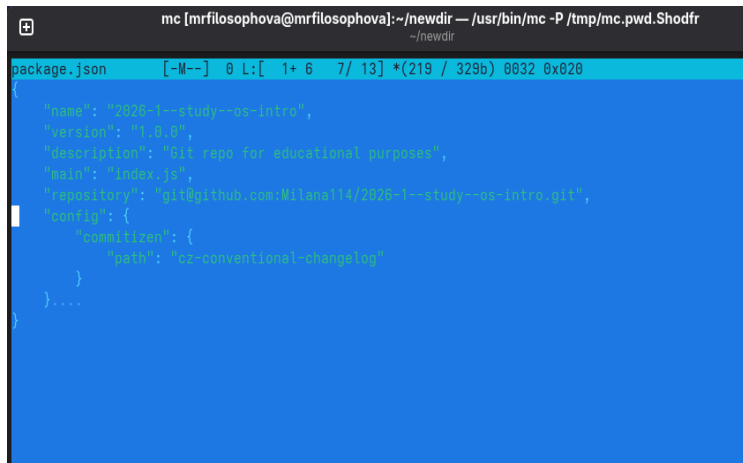
Рисунок 25: Распознавание клавиш



The screenshot shows the Midnight Commander (MC) interface. The title bar at the top reads "mc [mrphilosophova@mrphilosophova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.Shodfr" and the status bar below it shows "~/newdir". The main window displays the contents of a file named "package.json". The file's metadata bar shows "[----] 9 L:[1+11 12/ 15] *(371 / 384b) 0010 0x00A" and a search icon with "[*][X]". The file content is a JSON object with the following fields: "name", "version", "description", "main", "repository", "author", "license", and "config". The "config" field contains a "commitizen" object with a "path" field. The cursor is positioned at the end of the "commitizen" object.

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:Milana114/2026-1--study--os-intro.git",
  "author": "Milana114",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 26: Файл с текстом



The screenshot shows the Midnight Commander (MC) interface. The title bar at the top reads "mc [mrfilosophova@mrfilosophova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.Shodfr" and the current directory is "~/newdir". The file "package.json" is open and highlighted in cyan. The file's metadata is shown as "[-M--] 0 L: [1+ 6 7/ 13] *(219 / 329b) 0032 0x020". The content of the file is a JSON object for a package named "2026-1--study--os-intro", version "1.0.0". It includes a description, main file "index.js", repository URL, and configuration for commitizen.

```
package.json  [ -M-- ] 0 L: [ 1+ 6 7/ 13 ] *(219 / 329b) 0032 0x020
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:Milana114/2026-1--study--os-intro.git",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}....
}
```

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander

[illegible]

Рисунок 28: Копирование фрагмента

Работа с редактором Midnight Commander

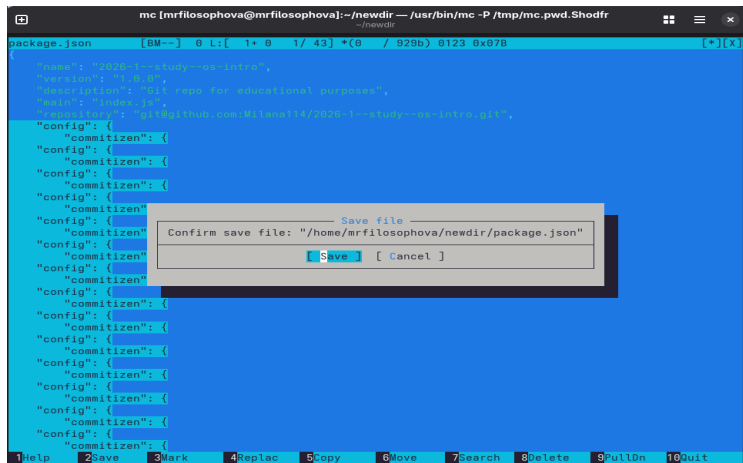


Рисунок 29: Сохранение

Работа с редактором Midnight Commander

```
mc [mrphilosophova@mrphilosophova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.Shodfr  
-/-newdir
```

```
{  
  "name": "2020-l--study--os-intro",  
  "version": "1.0.0",  
  "description": "git repo for educational purposes",  
  "main": "index.js",  
  "repository": "git@github.com:Milana147/2020-l--study--os-intro.git",  
  "config": {  
    "commitizen": {  
      "commitizen": {  
        "commitizen": {  
          "commitizen": {  
            "commitizen": {  
              "commitizen": {  
                "commitizen": {  
                  "commitizen": {  
                    "commitizen": {  
                      "commitizen": {  
                        "path": ".cz-conventional-changelog"  
                      }  
                    },  
                    .....  
                  }  
                },  
                .....  
              }  
            },  
            .....  
          }  
        },  
        .....  
      }  
    }  
  }  
}
```

```
1Help      2Save     3Mark      4Replac    5Copy      6Move      7Search    8Delete    9PullDn   10Quit
```

Рисунок 30: Отмена

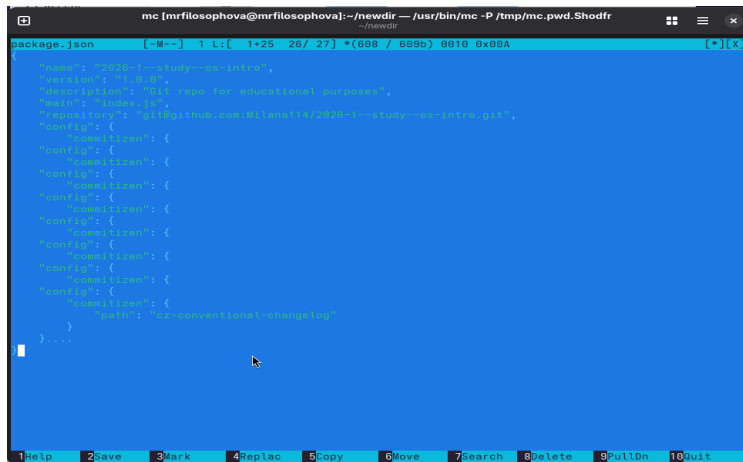


Рисунок 31: Переход в конец файла

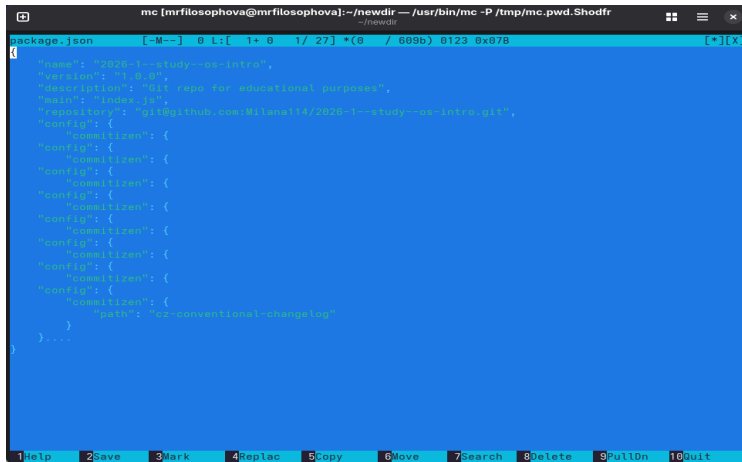
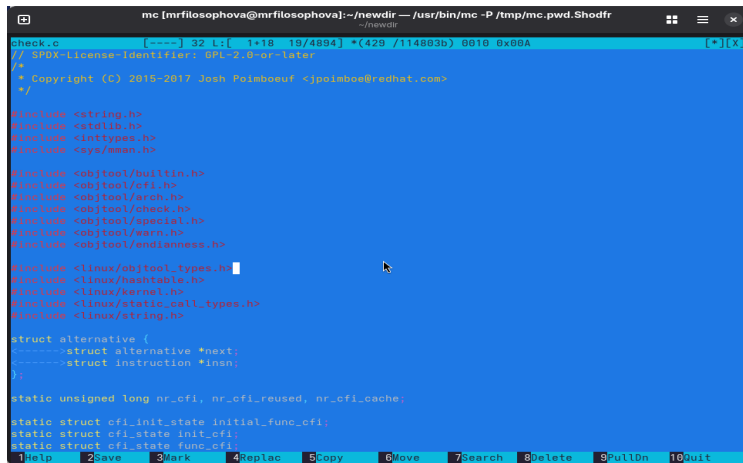


Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander



```
mc [mrfilosophova@mrfilosophova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.Shodfr
~/newdir
check.c [----] 32 L: [ 1+18 19/4894] *(429 /114883b) 0010 6x00A [*][X]
// SPDX-License-Identifier: GPL-2.0-or-later
/*
 * Copyright (C) 2015-2017 Josh Poimboeuf <jpoimboe@redhat.com>
 */

#include <string.h>
#include <stdlib.h>
#include <inttypes.h>
#include <sys/mman.h>

#include <objtool/builtin.h>
#include <objtool/cfi.h>
#include <objtool/arch.h>
#include <objtool/check.h>
#include <objtool/special.h>
#include <objtool/warn.h>
#include <objtool/endianness.h>

#include <linux/objtool_types.h>
#include <linux/hashtable.h>
#include <linux/kernel.h>
#include <linux/static_call_types.h>
#include <linux/string.h>

struct alternative {
    struct alternative *next;
    struct instruction *insn;
};

static unsigned long nr_cfi, nr_cfi_reused, nr_cfi_cache;

static struct cfi_init_state initial_func_cfi;
static struct cfi_state init_cfi;
static struct cfi_state func_cfi;

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

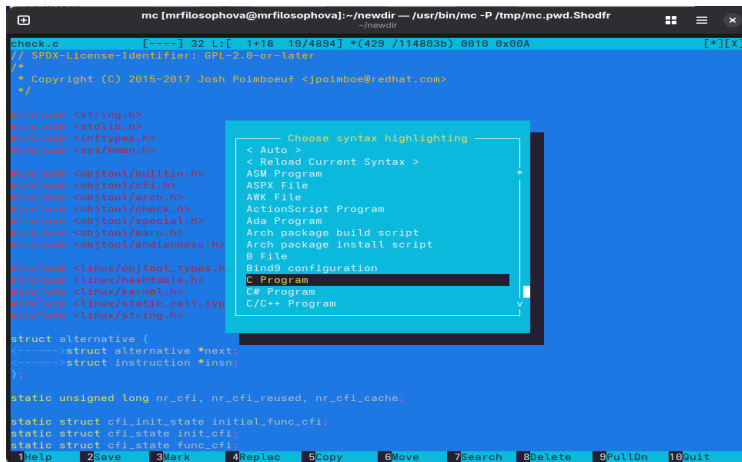


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.